

Northstar Distribution

August 2026 Reporting Reliability Audit

Synthetic demonstration - not a real client engagement.

PT Northstar Distribusi Nusantara is a fictional Indonesian-style B2B distributor. Observation: 8 September 2026, 09:00 WIB. Generated population: **demo** profile. All money is integer Indonesian rupiah (IDR). Percentages are rounded for display.

Decision required

Which August Revenue and Gross Margin should the CFO present for Board financial performance, and should the pack circulated on 2 September be reissued?

Reissue the financial-performance pack using Finance's closed P&L. Keep Sales' commercial measures alongside it with explicit labels. Both Revenue and Gross Margin are **CONFLICTING as reported; fully reconciled by the evidence**. The Board hybrid has mechanical errors and is unsuitable for the decision. This does not make Sales' commercial view wrong.

Surface	Revenue concept and amount	Gross margin concept and amount	Margin %
Finance, closed 7 September	RNMR: Rp 59,257,637,000	FGM: Rp 13,125,508,130	22.1499%
Sales, August deliveries	DSV: Rp 61,000,982,000	CM: Rp 12,200,196,400	20.0000%
Board, circulated 2 September	BR: Rp 60,035,534,525	BGM: Rp 11,234,748,925	18.7135%

Revenue verdict: CONFLICTING under a common label. RNMR is the appropriate August P&L basis. DSV remains a legitimate commercial delivery measure.

Gross Margin verdict: CONFLICTING under a common label. FGM is the appropriate P&L basis; CM remains legitimate for commercial performance at standard cost. Board Gross Margin mixes netted credits with unreversed delivery cost.

Measured business materiality

Relative to the closed Finance P&L, the circulated Board pack overstates Revenue by Rp 777,897,525 (1.3127%). Its Gross Margin difference is Rp -1,890,759,205, and its margin-rate difference is -3.4364 pp.

These are measured reporting differences, **not cash losses**. The August result feeds the second-half forecast and sales-incentive accrual; using an incompatible basis can change those decisions. No forecast loss or incentive overpayment has been quantified from the supplied evidence.

Immediate action: Finance Controller DA certifies the P&L values; the CFO appoints an accountable Board-pack owner; FP&A reissues the pack after close.

What the numbers actually mean

Finance - Recognized Net Merchandise Revenue (RNMR). Invoice accounting period equals invoice posting month; no accrual for unbilled deliveries. Merchandise is net of line discounts and recognized credit notes. VAT and freight are excluded. Actual Recognized COGS (ARC) uses actual cost effective at delivery, captured on invoice lines, less original-cost reversals on recognized credits. Finance Gross Margin (FGM) is RNMR minus ARC. ARC is Rp 46,132,128,870.

Sales - Delivered Sales Value (DSV). Calendar month of delivery, net of line discounts, excluding VAT, freight, cancellations and undelivered orders. Deliveries remain gross of returns; received returns are displayed separately. Commercial Margin (CM) subtracts standard cost effective at delivery, Rp 48,800,785,600. The August population is complete from the 1 September T+1 refresh. Subsequent returns do not revise historical delivered Revenue or Margin.

Board - as reported. Delivery-based sales minus credit-document totals using Finance's accounting period, restricted to documents present at the earlier run. Credit totals include VAT even though sales exclude it. Cost remains standard cost of all delivered goods, with no reversal for credits already deducted.

Revenue reconciliation

Each bridge starts with Finance RNMR and adds the following signed amounts. The categories distinguish legitimate definitions from snapshot differences and actual query errors; no unexplained residual remains. Each percentage is the signed line amount divided by starting Finance RNMR.

Finance RNMR to Sales DSV

Bridge line	Category	Signed change	% of RNMR
Delivery rather than invoice posting period	definition	Rp 627,000	0.0011%
Sales keeps deliveries gross of returns	definition	Rp 1,742,718,000	2.9409%

Endpoint: Sales DSV, Rp 61,000,982,000. Gross-of-returns reporting is intentional and legitimate for Sales' commercial context.

Finance RNMR to circulated Board Revenue

Bridge line	Category	Signed change	% of RNMR
Delivery rather than invoice posting period	definition	Rp 627,000	0.0011%
Credits posted after the Board snapshot	timing	Rp 872,945,500	1.4731%
VAT deducted on credits visible to Board	defect	Rp -95,674,975	-0.1615%

Endpoint: Board Revenue, Rp 60,035,534,525. Re-running the inherited SQL on the later horizon without the historical cutoff produces a different number; it does not reproduce the circulated pack.

Gross Margin reconciliation

Each bridge starts with Finance FGM. Revenue bridge lines are shown individually so their categories remain visible on the margin bridge. Each percentage is the signed line amount divided by starting Finance FGM.

Finance FGM to Sales CM

Bridge line	Category	Signed change	% of FGM
Delivery rather than invoice posting period	definition	Rp 627,000	0.0048%
Sales keeps deliveries gross of returns	definition	Rp 1,742,718,000	13.2773%
Actual cost: delivery rather than posting period	definition	Rp 3,397,160	0.0259%
Standard rather than actual cost	cost basis	Rp -1,314,142,820	-10.0121%
No reversal, consistent with gross-of-returns Sales	definition	Rp -1,357,911,070	-10.3456%

Endpoint: Sales CM, Rp 12,200,196,400.

Finance FGM to circulated Board Gross Margin

Bridge line	Category	Signed change	% of FGM
Delivery rather than invoice posting period	definition	Rp 627,000	0.0048%
Credits posted after the Board snapshot	timing	Rp 872,945,500	6.6508%
VAT deducted on credits visible to Board	defect	Rp -95,674,975	-0.7289%
Actual cost: delivery rather than posting period	definition	Rp 3,397,160	0.0259%
Standard rather than actual cost	cost basis	Rp -1,314,142,820	-10.0121%
Cost reversal on credits after the snapshot	timing	Rp -680,225,670	-5.1825%
Cost not reversed on credits already netted	defect	Rp -677,685,400	-5.1631%

Endpoint: Board Gross Margin, Rp 11,234,748,925.

Why the same cost amount receives different classifications

Total period COGS reversal R_{period} is Rp 1,357,911,070. It consists of R_{early} , Rp 677,685,400, plus R_{late} , Rp 680,225,670. $R_{period} = R_{early} + R_{late}$.

For Sales, not reversing this cost is a **definition difference**: DSV also keeps returns gross. For Board, the credits it already deducted require a corresponding cost reversal; omitting R_{early} is a **defect**. The later credits and their R_{late} reversals did not exist at the Board snapshot, so both are **timing**. The same credit-note membership split drives the revenue and cost attributions.

The Sales-to-Board differences are obtained by subtracting the Finance-to-Sales bridge from the Finance-to-Board bridge, category by category. The reconciliation therefore accounts for every pair of surfaces without adding another cause.

Evidence coverage and the two supporting findings

Evidence Coverage: 23 of 23 supplied assets inspected - the operational tables, reporting objects and job history in PostgreSQL, plus all seven supplied files. This is coverage of this bounded demonstration, not a general governance score. Inspection of all supplied assets does not establish an approved Board definition where no approval document exists.

Evidence group	What it establishes
Core operational tables	Order/delivery/invoice chronology; original prices, discounts and costs; warehouse returns; credit posting and period allocation. Payments corroborate lifecycle realism only.
Finance extract and finance_revenue.sql	Closed P&L values and component breakdown; Finance's posting-period and actual-cost basis.
Delivered-sales fact, dashboard view and sales_dashboard.sql	T+1 delivery population, gross-of-returns rule, standard cost and separately displayed return receipts.
Board snapshot, board_pack.sql and august_board_pack.csv	The circulated values, exact inherited logic and historical generated_at timestamp.
Job runs, comments and the three notes	Recurring close cadence, successful runs, Finance/Sales responsibilities and the absent Board approval/owner.

Supporting finding - freshness. Board ran on 2 September at 08:15. The back-dated August credits arrived in the 2-4 September batches, before Finance closed on 7 September at 17:30. Job history shows the same early-pack pattern across the generated history. Every run succeeded; this is a snapshot/close coordination problem, not a failed pipeline.

Supporting finding - ownership and definition. The November 2024 YP handover records a verbal agreement, says it was never written up and leaves the future owner as TBD. NS runs the pack, which does not establish accountable ownership. The Board table has no comments. Finance's notes describe its own policy but do not approve the pack. Sales' stale Revenue comment still says net of returns, while RW's early-2025 note and current SQL describe gross deliveries. These fragments establish ambiguity, not an agreed Board P&L definition.

Remediation and re-check

Sequence	Accountable action	Verification evidence
Before reissue	CFO chooses the Finance P&L basis and appoints the Board-pack accountable owner; DA certifies the closed values.	Written decision-context definitions and Finance sign-off.
Rebuild the pack	FP&A retires the hybrid pair. Present RNMR/FGM and DSV/CM separately with their explicit names, periods, cost bases and snapshots.	Reissued pack reconciles to Finance and Sales independently.
Before the next close	Assigned owner coordinates pack certification after Finance close; Sales Ops RW corrects the stale dashboard comment.	Agreed close-to-pack checklist and updated comment.
At the next re-check	Re-run surface queries and the bridges after close; preserve the previously circulated snapshot separately.	No unexplained residual; every new difference classified and signed off.

These are recommendations. The demonstration executes no remediation against source data and creates no recurring monitor.

Technical verification register

The following components come from core rows, independently of the reporting queries. Cross-period revenue legs are shown separately even though their net is small. Amounts are ex-VAT merchandise unless explicitly labelled otherwise.

Component	Generated amount
Gross invoice merchandise posted in August (G_post)	Rp 61,000,355,000
August credit merchandise (C_period)	Rp 1,742,718,000
Credit merchandise visible at Board run (C_early)	Rp 869,772,500
Credit merchandise after Board run (C_late)	Rp 872,945,500
August deliveries invoiced in September (D1)	Rp 2,514,536,000
July deliveries invoiced in August (D2)	Rp 2,513,909,000
Actual invoice cost in August (K_post)	Rp 47,490,039,940
Actual cost on D1 (K1)	Rp 1,955,399,080
Actual cost on D2 (K2)	Rp 1,958,796,240
Actual cost of August deliveries (K_act)	Rp 47,486,642,780
Standard minus actual delivery cost	Rp 1,314,142,820
VAT on visible credits (deducted by Board)	Rp 95,674,975

Materiality checks on the generated demo

Check	Observed	Contract threshold
C_period / RNMR	2.9409%	2.5%-4.0%
C_late / C_period	50.0910%	At least 35%
D1 / RNMR	4.2434%	At least 0.8%
D2 / RNMR	4.2423%	At least 0.8%
Absolute standard/actual difference / K_act	2.7674%	At least 1%
Finance margin level	22.1499%	15%-25%
Finance-Sales Revenue gap / RNMR	2.9420%	At least 0.5%
Finance-Board Revenue gap / RNMR	1.3127%	At least 0.5%
Sales-Board Revenue gap / RNMR	1.6292%	At least 0.5%
Finance-Sales margin-rate gap	2.1499 pp	At least 0.5 pp
Finance-Board margin-rate gap	3.4364 pp	At least 0.5 pp
Sales-Board margin-rate gap	1.2865 pp	At least 0.5 pp

These are demonstration-design materiality checks, not customer health scores.

Reproduce this audit

Use the repository's documented PostgreSQL setup and generate the **demo** profile from code and configuration into a clean local database. The committed sample uses the configured canonical seed; generated data and hidden QA truth are not committed. No service beyond local PostgreSQL is needed.

1. Compare the August Finance extract, Sales monthly totals across regions and Board CSV. Sum the Sales amounts before calculating its company-wide margin rate; do not average regional percentages.
2. Re-run `finance_revenue.sql` at the observation horizon. Its closed amounts reproduce unchanged. Re-run `board_pack.sql` with each historical run instant; `credit_notes.created_at` must be no later than `generated_at`. Using the horizon instead is a freshness experiment, not reproduction of the circulated pack.
3. Join invoices and merchandise lines to orders for D1/D2 and K1/K2. Recover standard and actual costs from their effective dates. Sum return quantities against the original invoice net unit price and captured actual cost.
4. Split August credit notes once at the Board `generated_at` timestamp. The visible set supplies `C_early`, its VAT and `R_early`; the complementary set supplies `C_late` and `R_late`. Rebuild the four bridges above.
5. Run the read-only Northstar re-check and repository tests. The re-check takes only database evidence and the requested period; it never reads the hidden manifest or uses generation targets to calculate the amounts.

```
python -m benchmark.scenario.reconcile --period 2026-08
python -m pytest
```

Technical source: `benchmark/scenario/reconcile.py` implements this bounded, read-only calculation. Surface queries and notes are supplied in `benchmark/data/evidence/`. Database evidence is the core, finance, analytics and management schemas. PostgreSQL is Northstar's canonical environment; the optional DuckDB mirror exists only for internal development and offline tests.

Complete supplied file inventory

`finance_revenue.sql`; `sales_dashboard.sql`; `board_pack.sql`; `august_board_pack.csv`; `finance_close_notes.md`; `sales_dashboard_notes.md`; `board_pack_handover.md`.

Boundaries and disclosure

This is one synthetic demonstration of Revenue and Gross Margin for a fictional company. It uses independently invented organizations, products, locations and role initials; no personal, employer or client records are used. The VAT rate is a fictional internal setting, not tax guidance. Historical behavior and the September tail support this specific management dispute.

There is no universal correct Revenue definition. Correctness depends on the stated decision and consistency of the chosen basis. The practical recommendation is to present Finance financial performance and Sales commercial performance separately, certify after Finance close, and retire the Board hybrid.